Andrew Hope

Curriculum Vitae

andhope@umich.edu — LinkedIn — (518)-769-4183

Education

B.S. Honors Physics and Mathematics
University of Michigan — 2027
Honors Thesis (in progress): *Debiasing cosmology for future LSS surveys with baryon constraints at small scales.*

Academic Awards and Honors

Walter Gray Fellowship (sole inaugural fellow)	University of Michigan Physics Dept. 2026
DESI Summer Fellowship (one of 15 inaugural awardees)	DESI Collaboration 2026
Honors Research Travel Grant	University of Michigan Honors Program 2026
Undergraduate Research Travel Award	University of Michigan Physics Dept. 2026
Ralph B. Bodine Scholarship	University of Michigan Physics Dept. 2025
James B. Angell Scholar	University of Michigan 2025
Otho Lyle Tiffany & Mary Lois Tiffany Fellowship	University of Michigan Physics Dept. 2024
Walter Grishkot Memorial Scholarship	Grishkot Foundation 2023

Research Experience

Debiasing Cosmology for Future LSS Surveys With Baryons
Advisors: Dragan Huterer, Uendert Andrade — Sept. 2025 - Present
Used Fisher forecasting to deliver concrete requirements for galaxy full-shape analyses so cosmology stays unbiased at smaller scales (higher wavenumber). Tested sensitivity of LSS simulation models when introducing priors across wavenumber bins with partial dependence functions. Plotted galaxy-galaxy power spectrum and angular power spectrum with varied baryon effects.

Bayesian Optimization for CCD Parameter Selection — *Walter Gray Fellow, FNAL*
Advisors: Alex Drlica-Wagner, Guillermo Fernandez-Moroni — June 2026 - Present
Optimal parameter selection (e.g. choosing pixel voltages), is a very costly effort when done manually. I wrote scripts to run CCD imaging, analysis, and a gaussian process based Bayesian optimization algorithm to automate finding best parameters. I also developed simulations to characterize importance of various sources of error in CCDs (amplifier noise, spurious charge, dark current, etc.) for uses in different science cases.

Piezoelectric Bender Fiber Positioner R&D
Advisor: Michael Schubnell — Sept. 2024 - Jan. 2026
Designed parts on Inventor and TinkerCAD, considering the anti-collision and small envelope constraints given for SPEC-S5 robots. 3D printed, hand-fabricated, and precision tested prototype fiber positioner robots that implement piezoelectric to the phi stage of the design from the DESI robots. Data analysis and control scripts with Python for the prototype positioners.

Fiber Positioner Robot Performance Testing — *SULI Intern, LBNL*
Advisors: Nick Wenner, Joe Silber, David Schlegel — June - Aug. 2025
Built test stand, assembled, soldered, and calibrated fragile fiber positioner robots. Tested physical properties of the robots, developed python control scripts for movement of multiple robots simultaneously or individually. Derived and wrote optimal correction-move scripts to improve precision from $\sim 77\%$ off-target after blind move to $\sim 4\%$ off-target post-correction. Correction scheme brings robots within 1 micron of error with repeated moves. Developed scripts to analyze multi-robot data under various movement tests. Created SQL system for storing all future robot testing results in the lab.

CMB-CIB Cross Correlation via Galactic Dust Maps
Advisors: Minh Nguyen, Dragan Huterer — Oct. 2023 - June 2024
Sorted bins and masked CMB data from Planck, cross-correlated power spectra of the CIB and CMB using CAMB and NaMaster. Plotted Mollweide projections of various CMB and CIB temperature maps.

Publications

Journal Publications

Hope, A., Huterer, D., Andrade, U. "Debiasing cosmology for future LSS surveys by understanding baryonic effects at high wavenumber." *In preparation.*

Conference Proceedings

Silber, J., Wenner, N., **Hope, A.**, Schlegel, D., et al. "Miniature fiber positioning stage with planetary gear configured in a new eccentric star ("ECSTAR") mode for 6.2 mm pitch robots" *In preparation.*

Wenner, N., Silber, J., Schlegel, D., **et al.** "Linear motion flexure ("R-FLEX") for miniature 6.2 mm pitch optical fiber positioning robots with polar (R-θ) kinematics" *SPIE Astronomical Telescopes & Instrumentation 2026, ArXiV.*

Schubnell, M., Fanning, T., **Hope, A.**, et al. "A Hybrid R–Theta Fiber Positioning Robot Using a Piezoelectric Bender for Radial Motion." *SPIE Astronomical Telescopes & Instrumentation 2026.*

Presentations

Oral Presentations

Hope, A., Uendert A., Huterer, D. "Debiasing cosmology for future LSS surveys by understanding baryonic effects at high wavenumber" *Fermilab New Perspectives Conference*, July 2026.

Poster Presentations

Hope, A., Fernandez-Moroni, G., Gamero, M., Drlica-Wagner, A. "Skipper CCD Parameter Optimization with Machine Learning" *APS Divisions of Particles and Fields*, July 2026.

Hope, A., Uendert A., Huterer, D. "Debiasing cosmology for future LSS surveys by understanding baryonic effects at high wavenumber" *APS Global Summit*, March 2026.

Hope, A., Wenner, N., Silber, J., Schlegel, D. "Lifetime testing of R-FLEX: flexure-based positioner robots for future massively parallel spectroscopic instruments" *APS Eastern Great Lakes Fall Meeting*, October 2025.

Hope, A., Wenner, N., Silber, J., Schlegel, D. "Lifetime testing of R-FLEX: flexure-based positioner robots for future massively parallel spectroscopic instruments" *Berkeley Lab WD&E Intern Poster Session*, August 2025.

Hope, A., Nguyen, M. "Cross-Correlation of the Cosmic Microwave Background and Cosmic Infrared Background" *Students in Applied Mathematics Symposium*, September 2024.

Hope, A., Nguyen, M. "Cross-Correlation of the Cosmic Microwave Background and Cosmic Infrared Background" *Undergraduate Research Opportunity Program Symposium*, April 2024.

Leadership, Teaching, and Outreach

Co-President, Society of Physics Students — Apr. 2026 - Present

Formerly **Vice President** (2025 - 2026), and **Tutoring Chair** (2024 - 2025). Lead the largest increase in student participation in UM SPS history, more than tripling average headcount. Scheduled professors for biweekly Speaker Series lectures. Planned resumé workshops, summer research application workshops, and research symposiums for undergraduate physics students. Held office hours 3x weekly for course advising, tutoring, and coding help.

Teaching Assistant, Physics 390: Modern Physics — Jan. - May 2025

Graded ~100 weekly written problem sets.

Undergraduate Instructor, Michigan Math and Science Scholars Program — July - Aug. 2024

Taught *Climbing the Distance Ladder: How Astronomers Survey the Universe*, an MMSS course for ~20 high school students. Led labs on N-Body simulations, VPython gravity simulations, and parallax calculations. Gave intro lectures on the solar system, vectors, special and general relativity, and the structure of a physics degree.

Extracurricular Activities

Athlete, University of Michigan Men's Rugby — 2024 - Present